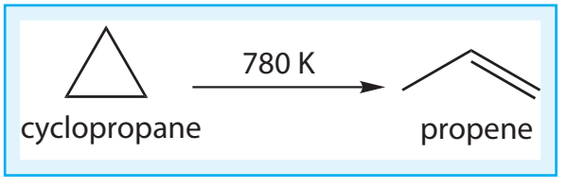
7.1.2 Average and instantaneous rate:

Let us understand the average rate and instantaneous rate by considering the isomerisation of cyclopropane.



The kinetics of the above reaction is followed by measuring the concentration of cyclopropane at regular intervals and the observations are shown below. (Table 7.1)

Table 7.1 Concentration of cyclopropane at various times during its isomerisation at 780K

Time (min)	[cyclopropane] (mol L^{-1})
0	2.00
5	1.67
10	1.40
15	1.17
20	0.98
25	0.82
30	0.69

$$\text{Rate of the reaction} = \frac{-\Delta\,[\text{cyclopropane}]}{\Delta t}$$

$$\text{The rate over the entire 30 min} = \frac{-(0.69-2)\ \text{mol}L^{-1}}{(30-0)\ \text{min}}$$

$$= \frac{1.31}{30} = 4.36 \times 10^{-2}\ \text{mol}L^{-1}\text{min}^{-1}$$

It means that during the first 30 minutes of the reaction, the concentration of the reactant (cyclo propane) decreases as an average of 4.36×10^{-2} mol L^{-1}each minute.

Let us calculate the average rate for an initial and later stage over a short period.

$$(\text{Rate})_{\substack{\text{initial}\\\text{stage}}} = \frac{-(1.4-2)}{(10-0)}$$

$$= \frac{0.6}{10} = 6 \times 10^{-2}\ \text{mol}\ L^{-1}\ \text{min}^{-1}$$

$$(\text{Rate})_{\substack{\text{later}\\\text{stage}}} = \frac{-(0.69-0.98)}{(30-20)}$$

$$= \frac{0.29}{10} = 2.9 \times 10^{-2}\ \text{mol}\ L^{-1}\text{min}^{-1}$$

From the above calculations, we come to know that the rate decreases with time as the reaction proceeds and the average rate cannot be used to predict the rate of the reaction at any instant. The rate of the reaction, at a particular instant during the reaction is called the instantaneous rate. The shorter the time period, we choose, the closer we approach to the instantaneous rate,

As $\Delta t \rightarrow 0$;

$$\frac{-\Delta[\text{cyclopropane}]}{\Delta t} = \frac{-d[\text{cyclopropane}]}{dt}$$

A plot of [cyclopropane] Vs (time) gives a curve as shown in the figure 7.2. Instantaneous rate at a particular instant 't' $\frac{-d\,[\text{cyclopropane}]}{dt}$ is obtained by calculating the slope of a tangent drawn to the curve at that instant.

In general, the instantaneous reaction rate at a moment of mixing the reactants (t = 0) is calculated from the slope of the tangent drawn to the curve. The rate calculated by this method is called initial rate of a reaction.

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